

Business Perspective

What are the business-related advantages of having the network you created?

The client-server network provides Company ABC with significant advantages that improve daily operations. By centralizing data and resources on a single server, employees can quickly access and share information, which reduces delays and eliminates repetitive tasks. Communication between departments becomes more efficient, and errors caused by misplaced or duplicated documents are minimized. The network also allows management to monitor activities, track progress, and make informed decisions based on up-to-date and accurate data. Overall, the system streamlines workflows, enhances productivity, and reduces operational costs.

What organizational expectations will be satisfied? Explain each.

Several organizational expectations are satisfied with this network. First, efficiency is promoted because employees can retrieve files instantly and coordinate work more effectively without unnecessary delays. Second, accountability is strengthened since the server logs user activities, showing who accessed or modified particular data. Third, data security and integrity are improved because information is stored centrally, regularly backed up, and protected from accidental loss or duplication. Finally, cost-effectiveness is achieved because the wired network minimizes hardware and maintenance expenses compared to a wireless setup while still providing reliable connectivity.

Will there be any downsizing, rightsizing, upsizing opportunities that may arise in the future? If so, what are the procedures that need to be done to make that opportunity possible?

Yes, the network provides flexibility for future adjustments. Upsizing can occur if the company grows, which involves adding more client computers or even a secondary server. This process requires connecting the new devices to the existing switch and configuring their IP addresses for network access. Rightsizing may involve redistributing workloads across servers to optimize performance, which can be done through load balancing or adjusting server roles. Downsizing, if ever necessary, would require removing unnecessary devices or reallocating resources, which is straightforward due to the centralized server structure. The network's scalability ensures that these changes can be implemented with minimal disruption and cost.

Technological Perspective

What are the technology-related advantages of having the network you created?

The client-server network provides centralized control over all resources, which improves efficiency in managing data, applications, and devices. Employees can share files and access applications quickly, reducing downtime and manual coordination. Communication between departments becomes faster through the LAN, and centralized backups help prevent data loss due to hardware failure or accidental deletion. Additionally, the network allows IT administrators to monitor and maintain systems more efficiently, making the technology more reliable and easier to scale as the company grows.

Are SNA, TCP/IP, or any other standards applicable to this scenario? If yes, elaborate and explain each standard.

Yes, multiple standards are applied. TCP/IP is used to ensure reliable data communication between the server and all client computers. Ethernet (IEEE 802.3) is used for wired connections, providing stable, high-speed network performance. The Client-Server architecture itself is a standard model where client computers request data or services from a central server, which manages and delivers the resources efficiently. These standards ensure the network operates consistently, securely, and interoperably across all devices.

Development Phases

- **Analysis:** The IT team assessed the number of users, the type of applications required, and the volume of data that would be stored and shared on the network.
- **Design:** A star topology was chosen, connecting all client computers to a central switch, which in turn connects to the server to optimize data flow and simplify troubleshooting.
- **Development:** Physical connections were established, IP addresses configured, and software installed on the server and client machines to support file sharing, printing, and application access.
- **Testing:** Network functionality was verified through file transfers, ping commands, and simulated work scenarios to ensure all devices communicated reliably.
- **Implementation:** The network was deployed for daily operations, allowing employees to access shared resources and applications seamlessly.
- **Maintenance:** Routine checks, software updates, backups, and security monitoring are performed to maintain performance and protect the network from failures or attacks.

Give any possible security issues that may arise at using client/server technologies in the given scenario.

Potential security issues include unauthorized access to the server, data theft, and malware introduced through removable media. There is also a risk of insider threats if employees misuse their access. These risks can be mitigated by implementing strong passwords, antivirus programs, firewalls, access restrictions, and regular backups. Physical security measures, such as restricting entry to the server room, are also essential to ensure the network remains secure and reliable.